

Java Training Class Topics Covered (Last 2 Days + Today)

Introduction to Java

- What is Java?
- Features of Java
- Platform Independence (Write Once, Run Anywhere)
- Applications of Java

Java Environment Setup

- JDK (Java Development Kit)
- Installing Java and Also VS-code
- Checking Java Version

Java Program Structure

- Java Boilerplate Code
- public Keyword
- class Keyword
- main() Method
- System.out.println()

Example:

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("Hello World");  
    }  
}
```

Creating and Running Java Programs

- How to create a .java file
- Naming conventions for Java files
- Compiling Java programs: javac Demo.java
- Running Java programs: java Demo

Variables and Literals

- What is a Variable?
- What is a Literal?
- Variable Declaration
- Variable Initialization
- Rules for Naming Variables

Example:

```
int age = 2(a*b);
```

Memory Management Basics

- How variables are stored in memory
- Variable allocation and storage

Data Types in Java

Primitive Data Types:

byte, short, int, long, float, double, char, boolean

Non-Primitive Data Types:

String, Array, Class, Object

Size and Range of Primitive Data Types

Data Type	Size	Example
byte	1 byte	byte age = 25;
short	2 bytes	short num = 1000;
int	4 bytes	int salary = 50000;
long	8 bytes	long population = 9000000L;
float	4 bytes	float price = 99.5f;
double	8 bytes	double pi = 3.14159;
char	2 bytes	char grade = 'A';
boolean	1 bit (logical)	boolean isJavaFun = true;

Example :

```
public class Main {  
    public static void main(String[] args) {  
        int age = 20;  
        double marks = 85.5;  
        char grade = 'A';  
        boolean passed = true;  
        System.out.println(age);  
        System.out.println(marks);  
        System.out.println(grade);  
        System.out.println(passed);  
    }  
}
```

Type Conversion

- **Implicit** Type Casting (Widening)
- **Explicit** Type Casting (Narrowing)

Example:

```
int a = 10;  
double b = a;
```

Lossy Conversion

- Data loss during type casting
- Narrowing conversion

Example:

```
double num = 10.75;  
int n = (int) num;
```

Input and Output

- Output using System.out.println()
- Introduction to Scanner Class
- Taking User Input

Example:

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");

        int age = sc.nextInt();

        System.out.println("Your age is: " + age);

    }

}
```

Comments in Java

- Single-line Comments : //
- Multi-line Comments : /* */

Keywords and Identifiers

- Java Keywords
- Identifiers
- Naming Rules

Escape Sequences

\n, \t, \", \\

Summary:

Students should understand Java Basics, JDK, Program Structure, Variables, Data Types, Type Casting, Operators, User Input, Comments, and Coding Standards.